

Comprehensive Guide to Generative AI

Structured revision notes for GenAI concepts, tools, frameworks, and workflows

This guide summarizes key course concepts including LLMs, prompting, RAG, agents, multi-agent systems, orchestration, vector databases, fine-tuning, and modern frameworks such as LangChain, LlamaIndex, LangGraph, AutoGen, CrewAI, BeeAI, FAISS, Pinecone, and Haystack.

Best way to revise

Start with the foundational concepts table, then memorize the RAG pipeline and multi-agent workflow. After that, compare tools by category: model development, deployment, retrieval, and infrastructure.

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Prepared as a clean, portable PDF for reading and revision.

Introduction

Generative AI (GenAI) has evolved from basic text and image generation to powering complex systems such as AI agents, enterprise automation, and reasoning engines.

This guide explores core concepts, tools, and frameworks for building modern GenAI applications, including RAG, multi-agent systems, prompt engineering, and cutting-edge libraries like LangGraph.

Whether you are developing chatbots, automation workflows, or knowledge systems, this guide provides a roadmap to the latest advancements. It also introduces additional terms not covered in the course videos. These terms are essential for enhancing your understanding of the course concepts.

Key idea

Modern GenAI applications usually combine an LLM with prompts, retrieval, memory or state, external tools, and orchestration logic.

Core GenAI Concepts and Terminologies

Foundational Concepts

Term	Definition	Examples / Use Cases
LLM	A type of AI model trained on vast amounts of text data to understand and generate human-like language.	GPT-01, Claude, LLaMA
Prompting	A technique for designing input instructions to guide LLM outputs.	'Write a summary in 3 sentences.' 'Answer as a cybersecurity expert.'
Prompt Templates	Reusable, structured prompts with placeholders for dynamic inputs.	'Explain {concept} like I am 5 years old.'
RAG (Retrieval-Augmented Generation)	Combines retrieval from external knowledge sources with LLM generation to enhance factual accuracy.	Answering questions with real-time data, such as using the RAG paper concept.
Retriever	A system component designed to fetch relevant information from a dataset or database.	Vector similarity search using FAISS or Elasticsearch.
Agent	An autonomous AI system that can plan, reason, and execute tasks using tools.	AutoGPT, LangChain Agents
Multi-Agent System	A framework in which multiple AI agents collaborate to solve complex tasks.	Microsoft AutoGen, CrewAI
Chain-of-Thought	A prompting technique that encourages models to decompose problems into intermediate steps.	'Let us think step by step...'
Hallucination Mitigation	Strategies to reduce incorrect or fabricated outputs from LLMs.	RAG, fine-tuning, prompt constraints
Vector Database	A database optimized for storing and querying vector embeddings.	Pinecone, Chroma, Weaviate
Orchestration	Tools to manage and coordinate workflows involving multiple AI components.	LangChain, LlamaIndex
Fine-tuning	Adapting pre-trained models for specific tasks using domain-specific data.	LoRA, QLoRA

Tools and Frameworks

Model Development and Deployment

Tool / Framework	Definition	Examples / Use Cases	Reference
Hugging Face	A platform hosting pre-trained models and datasets for NLP tasks.	Accessing GPT-2, BERT, Stable Diffusion	Hugging Face
LangChain	A framework for building applications with LLMs, agents, and tools.	Creating chatbots with memory and web search	LangChain
AutoGen	A library for creating multi-agent conversational systems.	Simulating debates between AI agents	AutoGen
CrewAI	A framework for assembling collaborative AI agents with role-based tasks.	Task automation with specialized agents	CrewAI
BeeAI	A lightweight framework to build production-ready multi-agent systems.	Distributed problem-solving systems	BeeAI
LlamaIndex	A tool to connect LLMs to structured or unstructured data sources.	Building Q&A; systems over private documents	LlamaIndex
LangGraph	A library for building stateful, multi-actor applications with LLMs.	Cyclic workflows, agent simulations	LangGraph

Retrieval and Infrastructure

Tool / Framework	Definition	Examples / Use Cases	Reference
FAISS	A library for efficient similarity search of dense vectors.	Retrieving top-k documents for RAG	FAISS
Pinecone	A managed cloud service for vector database operations.	Storing embeddings for real-time retrieval	Pinecone
Haystack	An end-to-end framework for building RAG pipelines.	Deploying enterprise search systems	Haystack

Advanced Prompting Techniques

Concept	Definition	Example
Few-Shot Prompting	Providing examples in the prompt to guide the model's output format.	Translate to French: 'Hello' -> 'Bonjour'; 'Goodbye' -> ____
Zero-Shot Prompting	Directly asking the model to perform a task without examples.	Classify this tweet as positive, neutral, or negative: {tweet}
Chain-of-Thought	Encouraging step-by-step reasoning.	First, calculate X. Then, compare it to Y. Final answer: ____
Prompt Chaining	Breaking complex tasks into smaller prompts executed sequentially.	Prompt 1: Extract keywords -> Prompt 2: Generate summary from keywords.

Prompting memory aid

Zero-shot means no examples. Few-shot means examples are included. Chain-of-thought encourages step-by-step reasoning. Prompt chaining divides a larger task into smaller sequential prompts.

Key Architectures and Workflows

RAG Pipeline

Step	Stage	What Happens
1	Retrieval	Query a vector database, such as Pinecone, for relevant context.
2	Augmentation	Combine the retrieved context with the user's prompt.
3	Generation	An LLM, such as GPT-4, produces the final answer.

Multi-Agent System

Component	Description
Agents	Specialized roles such as researcher, writer, or critic.
Orchestration	LangGraph for cyclic workflows, AutoGen for conversations, and similar frameworks.
Tools	Web search, code execution, API integrations, and other external capabilities.

Architecture memory aid

RAG improves factual grounding by adding retrieved context. Multi-agent systems divide work across specialized agents and coordinate them with orchestration tools.

References

Use these references to deepen your understanding of the concepts summarized in this guide.

- Retrieval-Augmented Generation (RAG) Paper
- Chain-of-Thought Prompting
- LangGraph Documentation
- CrewAI Documentation
- Prompt Engineering Guide

Quick Revision Checklist

- Define LLM, prompting, prompt templates, RAG, retriever, agent, vector database, orchestration, and fine-tuning.
- Explain how RAG works: retrieval, augmentation, and generation.
- Compare zero-shot, few-shot, chain-of-thought, and prompt chaining.
- Identify which tools are used for agents, orchestration, vector databases, and RAG pipelines.
- Describe how multi-agent systems use specialized agents, orchestration, and tools.